

Uncertainty and Sentiment in ARK Company Filings

Victor Chen | NetID: yc8027 | Fall 2026
github.com/victor4628/FRE7871-NLP

1. What I did

I study 1,536 filings from 91 companies in the ARK holdings universe, submitted during 2021-2025. I count negative and uncertainty words, compare proportions with TF-IDF, test language trends, and relate uncertainty to subsequent stock volatility and negative language to four-day excess returns. The strongest result is rising negative language within annual reports; incremental volatility prediction and pooled return prediction are not established.

2. Data construction

Security / company screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
Unresolved SEC ticker-to-CIK mapping	7	117
At least one 2021-2025 10-K/10-Q	24	93
Unique companies (GOOG and GOOGL count as one)	1	92

Table 1. Sequential filing screens	Removed	Remain
Downloaded security-filing records	0	1,702
Duplicate Alphabet accessions (retain GOOGL)	20	1,682
Parsed non-amended 10-K/10-Q	0	1682
At least 2,000 words (10-K) or 1,000 words (10-Q)	0	1682
Earliest filing per company-calendar quarter	14	1668
Usable day 0 and day -1 price at least \$3	108	1560
At least 60 return days before and after day 0	23	1537
Missing filing share count needed for size control	1	1536

The last row is an additional regression-data requirement: one JOBY filing has no usable share count, so size cannot be calculated. No further filing is lost to other controls or outcomes. All exhibits use this common sample. The earliest-filing rule uses submission quarter, across form types; it removes 14 distinct 10-Qs, not duplicate text.

Day 0 follows the assignment: convert acceptance time to Eastern; shift its date forward one day at or after 16:00; take the later of that date and the filing date, then the next available trading day. This moves 860 retained filings. Share counts are read from the same filing cover or a dated same-accession fact, never a later report. Prices and shares are aligned for splits.

Parsing keeps visible inline-XBRL narrative, removes hidden resources, and strips tables with more than 15% digits among non-space characters. The uppercase tokenizer retains words of at least two characters; there is no stemming or stopword removal. No filing failed parsing.

3. Word lists

The March 2026 Loughran-McDonald Master Dictionary 1993-2025 supplies 2,345 active Negative words and 297 active Uncertainty words; 40 occur in both lists. Ten Negative entries marked with a negative removal year are excluded. Negative words describe adverse conditions; uncertainty words qualify what may happen or what is known. Shared words and shared disclosure topics mean these are conceptually different but empirically related measures.

4. Method

For each filing, the proportional measure is list-word occurrences divided by all tokens. TF-IDF downweights repetition and words that occur in many documents. For term i in document j :

$$w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_i)$$

The weight is zero if tf is zero. Here tf is the term count, a is total tokens divided by distinct tokens, N is the number of documents and df is the number containing the term. A category score sums these weights. Natural logs reproduce the instructor checks: 0.8480, 0.2885, 0.5026. IDF is recomputed within each pooled or form-specific regression corpus.

For trends, each company receives equal weight within form and filing quarter. Separate 10-K and 10-Q series remove the seasonal form-mix problem. Aggregate regressions use time and quarter-of-year effects; Newey-West standard errors with four lags allow serial correlation. The within-company model adds company effects, comparing changes inside the same company. Company-clustered errors allow its repeated filings to be related.

For market outcomes, each language measure enters as a continuous regressor. Reported proportional-score coefficients are effects per 1 percentage-point increase in words (for example, 1.8% to 2.8%); reported TF-IDF coefficients are effects per 1 TF-IDF unit. The estimation code standardizes scores internally, then rescales the coefficients and standard errors to these original units. This does not change fitted values, t-statistics or p-values.

Both models include log size (day -1 price times filing shares), log mean daily dollar volume and stock-minus-SPY buy-and-hold return over [-60,-6], report type, company effects and calendar-quarter effects. Table 5 adds log prior volatility only in its second specification; Table 6 includes it. The code uses a 10-Q dummy with 10-K as reference, algebraically equivalent to a 10-K dummy with the intercept changed.

Realized volatility is daily-return sample SD multiplied by $\sqrt{252}$: 55 returns on [-60,-6] before and 60 on [+4,+63] after. The four-day return compounds days [0,+3] from the day -1 close, then subtracts the corresponding SPY return. No missing returns are filled. Outcome errors cluster by company.

5. What the measures are made of

Table 2. Filing-level summary statistics, by report type

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	381	2.252	0.421	1.975	2.269	2.530
10-K / Negative TF-IDF	381	162.114	55.855	117.780	148.951	198.860
10-K / Uncertainty %	381	1.935	0.241	1.790	1.953	2.114
10-K / Uncertainty TF-IDF	381	26.685	8.484	20.720	25.281	31.680
10-Q / Negative %	1,155	2.045	0.952	1.181	1.848	2.951
10-Q / Negative TF-IDF	1,155	78.481	65.438	23.966	54.937	126.426
10-Q / Uncertainty %	1,155	1.808	0.582	1.317	1.644	2.405
10-Q / Uncertainty TF-IDF	1,155	13.471	7.275	7.874	11.925	18.467

Proportions are percentages of all words; TF-IDF is the sum of term weights. P25 and P75 are percentiles.

Table 3. Thirty most frequent words on each list

Each share is the word's fraction of all occurrences on its own list.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.29%	MAY	34.79%
2	ADVERSELY	4.19%	COULD	18.35%
3	LOSSES	3.26%	RISK	5.05%
4	CLAIMS	3.08%	RISKS	4.53%
5	ADVERSE	2.79%	BELIEVE	2.95%
6	AGAINST	2.48%	APPROXIMATELY	2.65%
7	LITIGATION	1.99%	ASSUMPTIONS	1.91%
8	UNABLE	1.97%	INTANGIBLE	1.70%
9	HARM	1.91%	POSSIBLE	1.32%
10	FAILURE	1.83%	UNCERTAINTIES	1.16%
11	FAIL	1.35%	FLUCTUATIONS	1.14%
12	IMPAIRMENT	1.27%	MIGHT	1.09%
13	NEGATIVELY	1.17%	UNCERTAIN	1.03%
14	DIFFICULT	1.16%	ANTICIPATED	0.98%
15	PENALTIES	1.16%	VOLATILITY	0.94%
16	NEGATIVE	1.04%	DEPEND	0.94%
17	DAMAGES	0.94%	PREDICT	0.94%
18	DECLINE	0.94%	UNCERTAINTY	0.89%
19	DELAYS	0.93%	DIFFER	0.86%
20	RESTATED	0.91%	ANTICIPATE	0.80%
21	LIMITATIONS	0.88%	EXPOSURE	0.79%
22	DELAY	0.83%	CONTINGENT	0.78%
23	VOLATILITY	0.79%	VARIABLE	0.71%
24	CHALLENGES	0.77%	DEPENDS	0.68%
25	FINES	0.73%	PENDING	0.65%
26	HARMED	0.71%	DEPENDENT	0.58%
27	DISRUPTIONS	0.70%	CONTINGENCIES	0.56%
28	INVESTIGATIONS	0.69%	PROBABLE	0.52%
29	BREACH	0.66%	ASSUMED	0.52%
30	INFRINGEMENT	0.62%	VARY	0.49%

Q1. What is each measure actually made of? The top ten words contribute 28.8% of Negative and 74.4% of Uncertainty occurrences. Uncertainty therefore measures much recurring conditional language, not just fresh managerial doubt. In PayPal's 2024-filed 10-K, 173/232 (74.6%) uncertainty-containing sentences in Risk Factors also appear in its previous annual report after token normalization (minimum 12 words). Repeated warnings concern cyberattacks, fraud and business interruptions. This is direct evidence of retained risk disclosure in that case; it does not show that all uncertainty language across the sample is copied.

Q2. Are sentiment and uncertainty measuring different things? Pooled correlations are 0.843 (proportions) and 0.918 (TF-IDF); within 10-K they are 0.725/0.849, and within 10-Q 0.849/0.919. These are largely overlapping disclosure measures, not two independent confirmations. PayPal's 10-K filed 2024-02-08 (fiscal year ended 2023-12-31) illustrates a difference: Negative is 2.804% (89th percentile of 10-Ks), Uncertainty 1.790% (25th). Its discussion of transaction losses, credit-loss provisions and fraud protection raises Negative counts; some words describe ordinary business risks, rather than new bad news. Its four-day SPY excess return is -7.07%; annualized volatility falls from 37.94% to 25.14%. Those are descriptive case outcomes, not evidence that the wording caused them.

6. Trends, 2021-2025

Figure 1 separates report types and averages companies equally. Table 4 further adjusts for company identity and quarter-of-year seasonality. These corrections address changing form and company composition before interpreting trends.

Figure 1. Negative and uncertainty measures by filing quarter; VIX on right axes

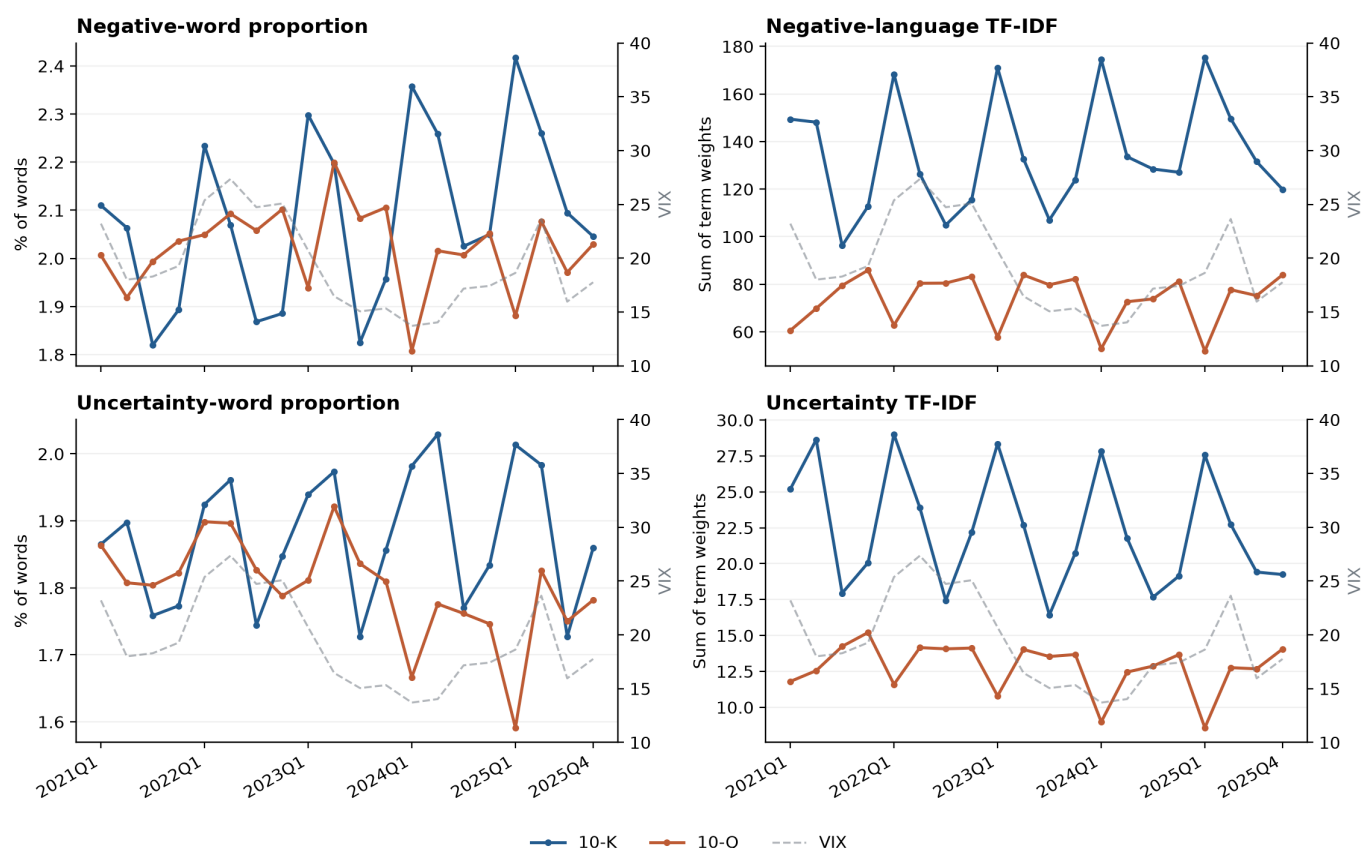


Table 4. Time-trend coefficient and t-statistics

Form / measure	Aggregate / year	OLS t	NW t	Within / year	Within t	Within p
10-K / Negative %	0.062	9.36	13.35	0.065	7.12	<0.001
10-K / Negative TF-IDF	2.797	3.20	5.76	2.616	3.77	<0.001
10-K / Uncertainty %	0.018	3.52	3.20	0.025	5.16	<0.001
10-K / Uncertainty TF-IDF	-0.194	-1.01	-1.51	-0.029	-0.26	0.797
10-Q / Negative %	-0.010	-0.85	-0.88	-0.005	-0.36	0.719
10-Q / Negative TF-IDF	-1.374	-1.89	-2.03	-1.504	-1.26	0.212
10-Q / Uncertainty %	-0.029	-2.88	-3.05	-0.020	-1.97	0.052
10-Q / Uncertainty TF-IDF	-0.516	-3.97	-4.64	-0.482	-3.46	<0.001

Dependent variable: the named language measure. Predictor: linear time in years. Proportional slopes are percentage points/year; TF-IDF slopes are term-weight units/year. OLS and NW t-statistics refer to the aggregate model; within t and p use company-clustered errors.

Q3. Did either measure trend over 2021-2025? I lead with the required within-company model because it removes stable differences between companies, not because it produces a preferred sign. Its similarity to the aggregate result supports the direction after that correction. Annual Negative rises 0.065 percentage points/year ($t=7.12$) and 2.616 TF-IDF units/year ($t=3.77$). Annual Uncertainty rises proportionally (0.025 points/year, $t=5.16$) but not in TF-IDF ($t=-0.26$). Quarterly Negative has no clear trend; quarterly Uncertainty falls -0.020 points/year ($t=-1.97$) and -0.482 TF-IDF units/year ($t=-3.46$). This supports some real within-company language change rather than a trend created entirely by composition; it does not prove management became more pessimistic or uncertain.

7. Uncertainty, volatility and returns

Table 5A. Does uncertainty predict subsequent volatility? Full sample

Dependent variable: log annualized volatility on [+4,+63]. Main independent variable: continuous uncertainty, computed separately as proportion or TF-IDF. Columns distinguish whether prior volatility is omitted or included. All columns include size, dollar volume, prior excess return, report type, company and calendar-quarter effects.

Variable / statistic	Proportion / Without prior-vol control	Proportion / With prior-vol control	TF-IDF / Without prior-vol control	TF-IDF / With prior-vol control
Uncertainty coefficient	0.0229	0.0005	0.0035	0.0014
Uncertainty clustered SE	0.0299	0.0250	0.0027	0.0019
Prior-volatility coefficient	-	0.3698	-	0.3682
Prior-volatility clustered SE	-	0.0964	-	0.0956
Uncertainty p-value	0.446	0.984	0.193	0.443
Filings / companies	1,536 / 91	1,536 / 91	1,536 / 91	1,536 / 91

Proportion coefficients are per 1 percentage point of all words; TF-IDF coefficients are per 1 term-weight unit. Standard errors cluster by company.

Q4. Does uncertainty language predict volatility? Adding prior volatility changes the proportional coefficient from 0.0229 to 0.0005 log points per percentage point, and TF-IDF from 0.0035 to 0.0014 log points per unit. The declines are 97.8% and 59.7%, consistent with prior volatility capturing pre-existing risk. However, the no-prior-volatility coefficients are already imprecise after the other required controls and fixed effects, so this sample does not reproduce the anticipated large unadjusted relation; the controlled estimates show no incremental predictive content.

Table 6. Does negative language predict four-day excess returns?

Outcome: stock-minus-SPY buy-and-hold return on [0,+3], in percentage points. Predictor: continuous Negative proportion or TF-IDF. Controls: log size, log dollar volume, prior excess return and volatility, plus report type, company and quarter effects. Effects are per 1 percentage point or 1 TF-IDF unit.

Negative-language predictor	Coefficient	Clustered SE	t	p	80% MDE
Negative % (+1 pp)	-0.750	0.700	-1.07	0.287	1.980 pp
Negative TF-IDF (+1 unit)	-0.012	0.010	-1.18	0.241	0.028 pp

N=1,536 filings and 91 company clusters. Approximate 80%-power minimum detectable effect (MDE) = (t critical at 5%, two-sided + 0.842) x clustered SE: 1.980 return percentage points per 1 proportion point and 0.028 return percentage points per TF-IDF unit. These are precision diagnostics for detecting effects, not measured test power.

Q6. Which of your results do you believe? I have the most confidence in the direction of the annual-report Negative trend within this selected sample. Proportion and TF-IDF both rise inside companies, with t-statistics of 7.12 and 3.77, and the aggregate estimates point the same way. This supports a change in filing language among surviving ARK issuers. It does not establish that their underlying businesses became more negative, because dictionary counts also capture recurring legal and accounting language.

I do not believe the evidence supports a general rise in Uncertainty. The annual proportion rises but annual TF-IDF does not, whereas both quarterly estimates fall; the measures are reacting differently to common hedging words and less common terms. For volatility, I believe the strong persistence in volatility and the absence of clear incremental information in the pooled language scores: after prior volatility is added, both uncertainty estimates are imprecise, with 95% intervals of [-0.049, 0.050] log points per proportion point and [-0.002, 0.005] per TF-IDF unit. The isolated controlled 10-K TF-IDF p-value in section 8 is exploratory because it is one of several form/weight specifications and no formal interaction test shows that 10-K and 10-Q effects differ. The return coefficients are negative, but their confidence intervals include zero and the MDEs are large. I therefore treat the return result as inconclusive, rather than evidence of no effect.

8. 10-K versus 10-Q

Table 5B. Uncertainty and subsequent volatility, by filing type. Table 5A estimates one common slope from all filings; Table 5B estimates the same model separately for 10-K and 10-Q. Every row uses Uncertainty; headings state the weighting and prior-volatility choice. Other controls are unchanged, and IDF is recomputed within each form.

Form	Uncertainty weighting	Without prior-vol control: coefficient / p	With prior-vol control: coefficient / p	N
10-K	Proportion (+1 pp)	-0.126 / 0.819	-0.132 / 0.796	381
10-K	TF-IDF (+1 unit)	0.038 / 0.062	0.033 / 0.041	381
10-Q	Proportion (+1 pp)	0.080 / 0.151	0.014 / 0.714	1,155
10-Q	TF-IDF (+1 unit)	0.008 / 0.062	0.003 / 0.280	1,155

Q5. Do 10-Qs behave like 10-Ks? The forms do not behave identically. Median length is 22,002 words for 10-Q versus 61,691 for 10-K. Negative-proportion SD is 0.952 versus 0.421 percentage points; Uncertainty SD is 0.582 versus 0.241. A shorter denominator can make a proportion more sensitive to a few changed words; repeated templates can instead suppress within-company variation. The observed overall variance is higher for 10-Q, but length alone does not explain it.

Annual reports contain a fuller discussion of business risks, so they are a plausible source of more company-specific text. That is a hypothesis, not a result of having more pages. Only annual Uncertainty TF-IDF is significant in the controlled form-specific volatility test; annual proportions and both quarterly measures are not. Different significance levels do not prove the forms differ. Table 4 similarly shows an annual Negative increase and a quarterly Uncertainty decline, not a common form-independent trend.

A 10-Q can arrive close to an earnings release, which may reveal information first or fall within the event window. Earnings-release timestamps are not matched here. Therefore, even a significant filing-date return association would not isolate the incremental effect of the 10-Q text.

9. Limitations

The 2026 holdings snapshot selects survivors; failed or exited former holdings are missing, and only 71/91 retained issuers appear in every filing year. Thus trends among survivors need not describe the historical ARK portfolio or the broader market. Only 20 filing quarters identify the aggregate time trend, so Newey-West inference is approximate in a short series.

The dictionary does not read context, negation or whether a sentence is new. The concentration of Uncertainty counts and the PayPal repetition result show that template language can drive scores. The regression tables use the full analysis corpus to calculate TF-IDF document frequencies, so those scores are retrospective. The portfolio extension recomputes IDF using only filings public at each date, but it still inherits the report's filtered estimation sample. Daily event timing cannot separate filing text from simultaneous earnings news, and the multi-class share aggregation used for size is an approximation.

The current report includes aggregate and within-company trends; full-sample and form-specific volatility models with and without prior volatility; and SPY and ARKK return benchmarks. Multiple related tests raise the chance of an isolated small p-value. With ARKK, Negative coefficients are -0.535 per proportion point ($p=0.418$) and -0.007 per TF-IDF unit ($p=0.443$); the imprecise pooled return conclusion persists.

Earlier exploratory work also ran no-control, size-only, prior-volatility-only and size-plus-volatility outcome models, an issuer/time-effects sensitivity, a shell-company screen, a temporary sample without the \$3 cutoff, and a return-quintile plot. The September 7 instructions superseded those choices. They are disclosed in AI_USE.md and were not used to choose the reported conclusion. The PayPal repetition check remains one case, not a corpus-wide duplication estimate.

10. What I would do next

The single change that would most improve the test is reconstructing point-in-time ARK membership from dated holdings archives, then including each company only during periods when an ARK fund actually held it. This would restore firms later sold, delisted or failed and directly test whether the conclusions survive outside today's selected survivors. The cost is reconciling ticker, name and CIK changes and recovering filings plus delisting-adjusted prices; missing archives may also shrink the usable sample.

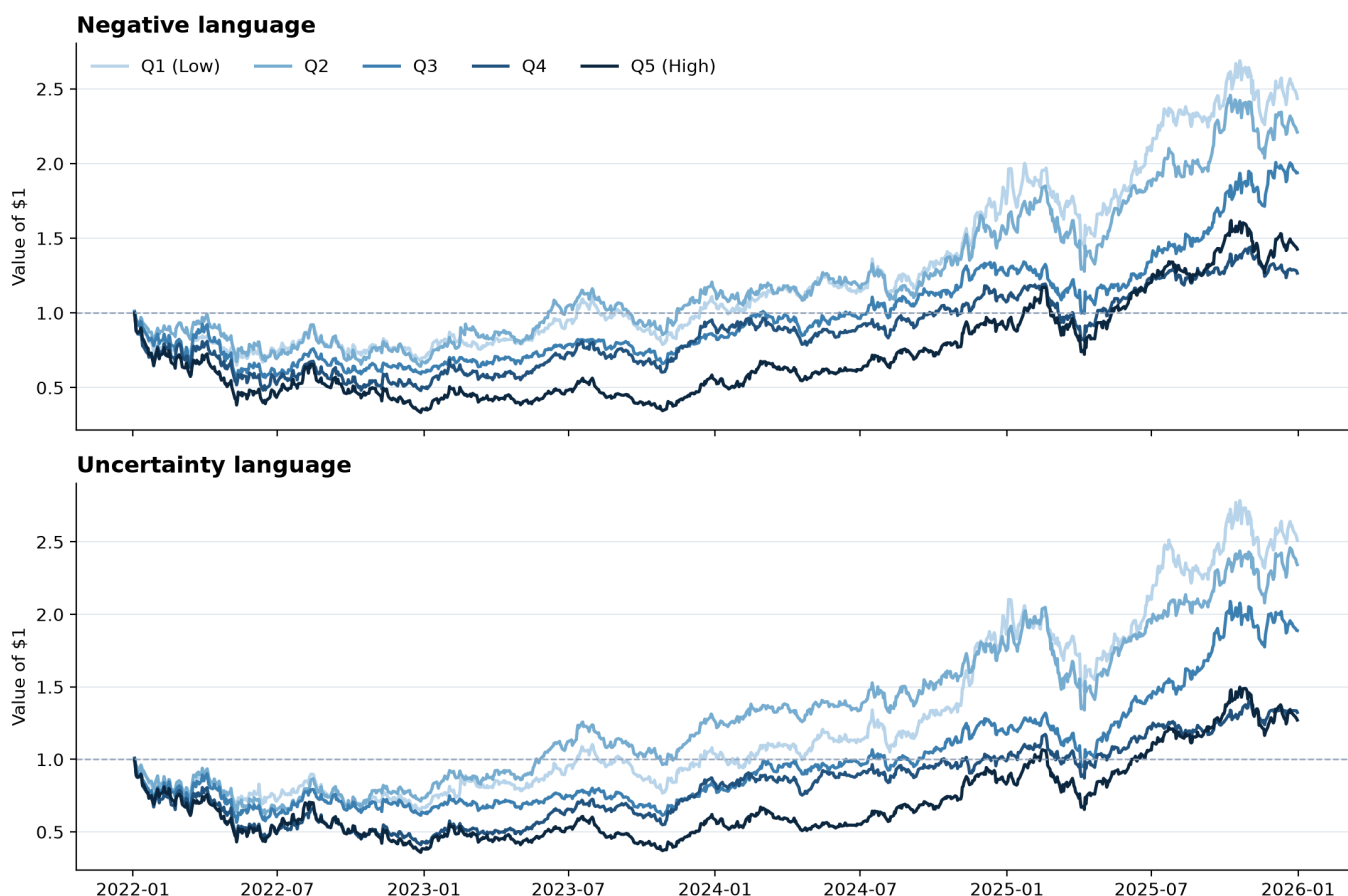
I would also extend the portfolio history and test the apparent rotation directly: does Q5 catch up after Q1-Q5 becomes unusually wide, and do reversals line up with earnings seasons or market-volatility regimes? I would evaluate the rule after transaction costs and in a genuinely out-of-sample period. The present four-year chart is too short to treat midyear or year-end movements as a seasonal strategy.

11. Exploratory portfolio sorts

As an extension of the report results, I update each issuer's language score when a retained filing becomes public and sort the available companies into five equal-weight portfolios. TF-IDF document frequencies use only filings public by that time. Scores are first ranked against earlier filings of the same form and then against earlier filings from the same calendar filing quarter, allowing 10-K and 10-Q observations to enter one ranking while controlling their systematic level differences. Q1 contains the lowest language scores and Q5 the highest. Portfolios rebalance at the first eligible market open and use adjusted raw stock returns.

Figure 2. Filing-language quintile portfolios, 2022-2025

Filing-language quintile portfolios - assignment report sample



Q1 has the lowest adjusted point-in-time TF-IDF percentile and Q5 the highest. Equal-weight portfolios rebalance at eligible filing-event opens; returns are adjusted raw stock returns.

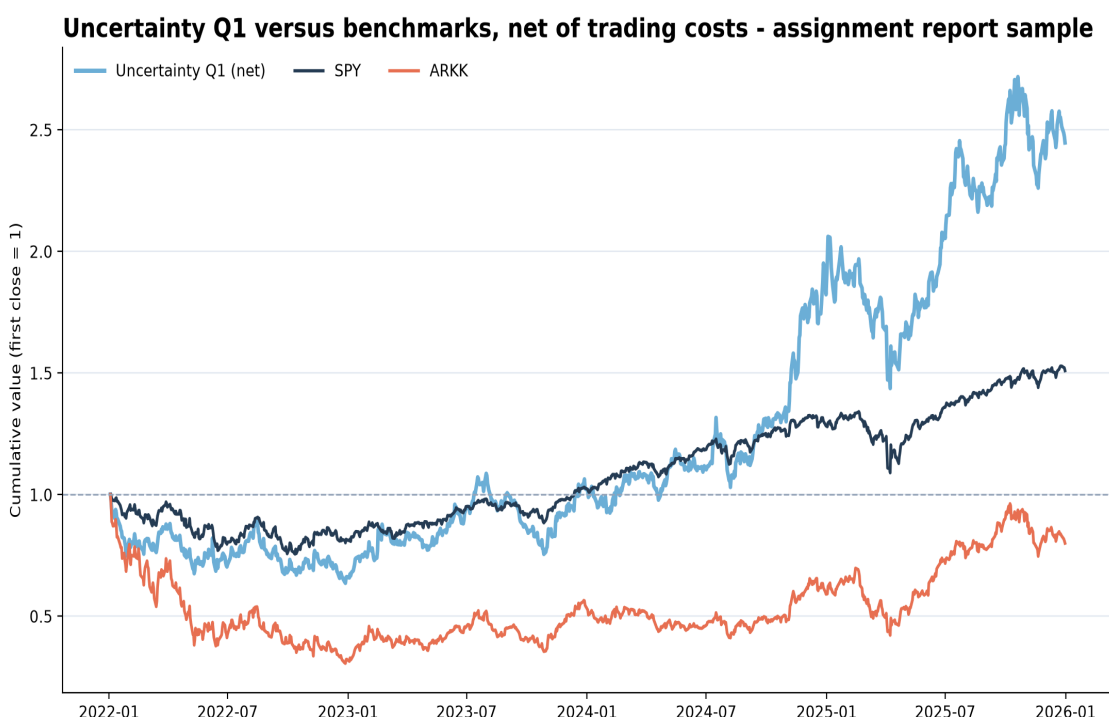
Negative Q1 ends at \$2.44 and Q5 at \$1.43; Uncertainty Q1 ends at \$2.51 and Q5 at \$1.27. The ordering is strongest between the low- and high-language groups, but it is not perfectly monotonic across all five portfolios. These are descriptive extensions of the same 1,536-filing report sample. The sample was selected for the assignment regressions, and transaction costs are omitted, so the figure is not a clean out-of-sample trading test.

Table 7. Quintile portfolio performance

Measure	Portfolio	Total return	Annualized return	Sharpe ratio	Maximum drawdown
Negative	Q1	143.6%	25.1%	0.82	-35.9%
Negative	Q2	121.0%	22.1%	0.73	-35.9%
Negative	Q3	93.8%	18.1%	0.65	-44.2%
Negative	Q4	26.6%	6.1%	0.35	-52.1%
Negative	Q5	42.7%	9.3%	0.42	-66.7%
Uncertainty	Q1	151.0%	26.0%	0.82	-36.1%
Uncertainty	Q2	134.3%	23.9%	0.79	-38.4%
Uncertainty	Q3	88.8%	17.3%	0.63	-43.8%
Uncertainty	Q4	32.4%	7.3%	0.38	-59.1%
Uncertainty	Q5	27.1%	6.2%	0.36	-64.1%

Returns cover January 3, 2022 through December 31, 2025. Annualized return compounds the ending value over 1,003 trading days. Sharpe ratios use daily returns, 252 trading days and a zero risk-free rate. Transaction costs are omitted.

Figure 3. Uncertainty Q1 versus SPY and ARKK

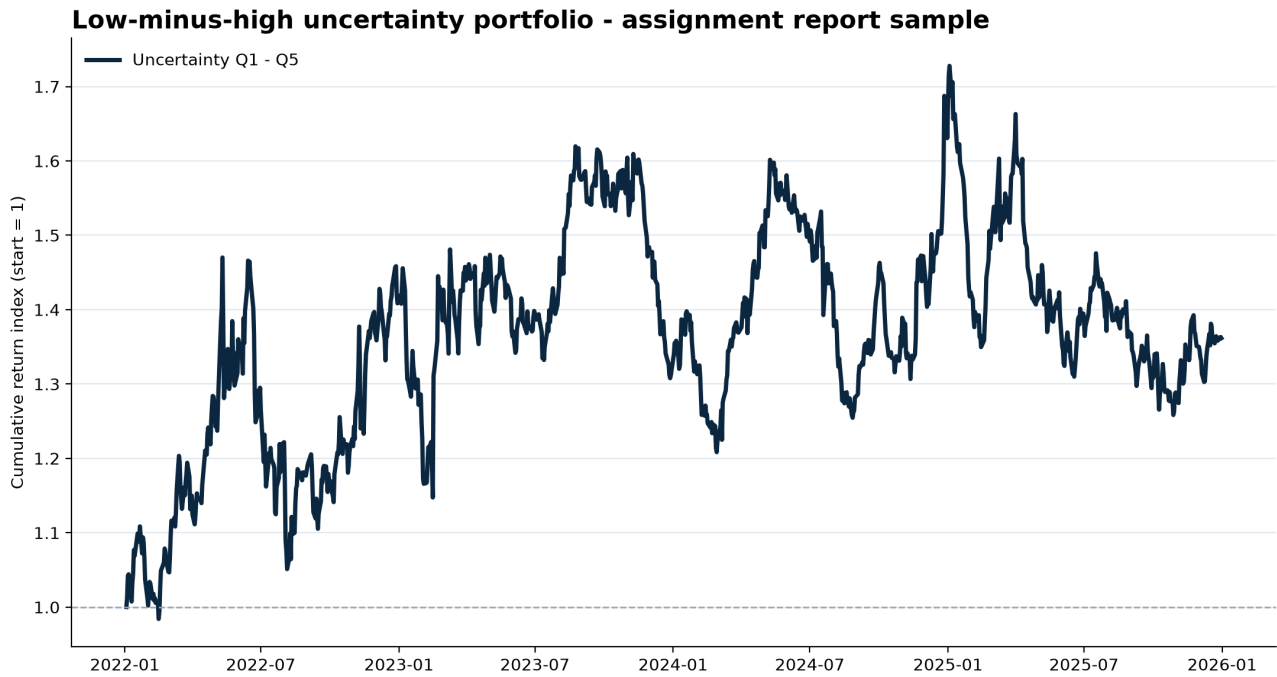


Uncertainty Q1 contains the lowest adjusted TF-IDF percentiles and deducts 10 basis points per 100% of one-way turnover. All series are normalized at the first portfolio close; SPY and ARKK use adjusted close prices.

Uncertainty Q1 contains the lowest 20% of adjusted uncertainty scores. I deduct 10 basis points for each 100% of one-way turnover; the five-portfolio results in Figure 2 and Table 7 remain gross of costs. The comparison normalizes each series to one at the first portfolio close. Uncertainty Q1 outperforms SPY and ARKK in this selected sample, but the result still inherits the sample-selection limitations stated above.

Long-short results

Figure 4. Low-minus-high Uncertainty portfolio



Long Q1 (lowest adjusted Uncertainty TF-IDF percentile) and short Q5 (highest). Daily return is Q1 minus Q5. No benchmark and no trading costs.

When I first saw the fairly clear ordering across the five Uncertainty portfolios, I expected a simple long-short strategy to perform extremely well. Figure 4 shows why that conclusion is too strong. The strategy is profitable overall, but its path is unstable and includes repeated reversals. Low-Uncertainty companies do not outperform continuously; high-Uncertainty companies periodically catch up. The chart suggests that some of these catch-up episodes occur around parts of midyear and near year-end, but I do not find a strong or repeatable seasonal rule in this four-year sample.

Figure 5. Low-minus-high Negative portfolio



Long Q1 (lowest adjusted Negative TF-IDF percentile) and short Q5 (highest). Daily return is Q1 minus Q5. No benchmark and no trading costs.

Negative language shows a similar pattern and perhaps a clearer change in trend. From 2022 through much of 2024, low-Negative companies generally outperform high-Negative companies. After 2024, the spread reverses and gives back a large part of its earlier gains. The five-portfolio ordering therefore does not translate into a stable long-short return in every subperiod.

Table 8. Latest companies at each language-score extreme

The table reports the three lowest and three highest current adjusted percentiles for each measure as of December 31, 2025. Percentiles use the latest retained filing for each company.

Measure	Extreme	Ticker	Company	Latest filing	Percentile
Negative	Lowest	NFLX	NETFLIX INC	2025-10-22	1.4
Negative	Lowest	HEI	HEICO CORP	2025-12-22	2.4
Negative	Lowest	AUR	Aurora Innovation, Inc.	2025-10-28	6.1
Negative	Highest	CMPS	COMPASS Pathways plc	2025-11-04	99.6
Negative	Highest	HOOD	Robinhood Markets, Inc.	2025-11-06	97.6
Negative	Highest	COIN	Coinbase Global, Inc.	2025-10-30	97.1
Uncertainty	Lowest	HEI	HEICO CORP	2025-12-22	3.9
Uncertainty	Lowest	TDY	TELEDYNE TECHNOLOGIES INC	2025-10-24	5.2
Uncertainty	Lowest	AUR	Aurora Innovation, Inc.	2025-10-28	6.4
Uncertainty	Highest	NRIX	Nurix Therapeutics, Inc.	2025-10-09	97.6
Uncertainty	Highest	SDGR	Schrodinger, Inc.	2025-11-05	96.2
Uncertainty	Highest	CMPS	COMPASS Pathways plc	2025-11-04	95.1

The endpoint lists are screening results, not buy or sell recommendations. A low score can reflect concise or repetitive disclosure, while a high score can reflect the company's business model or reporting detail rather than new information.

Sources: Loughran and McDonald (2011), Journal of Finance 66, 35-65; instructor assignment sheet and repository instructions; SEC EDGAR; Yahoo Finance. Saved calculations, full regression coefficients and filing-text comparisons are in the notebook. AI assistance is disclosed in AI_USE.md.